

Customer Churn Prediction — Stage 3: ANN Refinement and Insights

1. Defined Architecture of ANN Model

The final artificial neural network (ANN) used for churn prediction is a feed-forward network made up of an input layer, three hidden layers, and a single output layer. Each part of the design was chosen deliberately, and validated against four alternative configurations, so that the network's structure would be optimised for predictive performance rather than picked arbitrarily.

Input Layer

The input layer consists of 20 neurons, one for each feature in the full Stage 2 feature set (covering demographic details, service subscriptions, contract terms, and billing information). Using the complete feature set — rather than a reduced subset — gave the model the richest signal to learn from, and this proved to be a key factor in its predictive performance.

Hidden Layers

Three hidden layers taper progressively from 64 to 32 to 16 neurons. Each layer follows the same repeating block: a Dense layer with ReLU activation, followed by Batch Normalisation, followed by Dropout. This combination keeps training stable and reduces the risk of the network overfitting to the training data.

Layer	Neurons	Activation	Dropout	Notes
Hidden 1	64	ReLU	0.30	He-normal initialisation; L2=0.001
Hidden 2	32	ReLU	0.20	He-normal initialisation; L2=0.001
Hidden 3	16	ReLU	0.00 (none)	He-normal initialisation; L2=0.001

All three hidden layers apply an L2 weight penalty of 0.001 and use He-normal weight initialisation, which is well suited to ReLU-based networks. The progressively narrowing shape (64 → 32 → 16) allows the network to first learn broad patterns in the data, then gradually compress that information into a more concentrated churn signal.

Output Layer

The output layer is a single neuron with a sigmoid activation function, producing a churn probability between 0 and 1. This probability is converted into a churn / no-churn prediction using a threshold of 0.52, which was tuned on the validation set rather than fixed at the conventional 0.5, to better balance precision and recall for this business problem.

Optimising the Architecture for Predictive Performance

To make sure the architecture above was genuinely the strongest option rather than just the first one tried, it was benchmarked against four alternatives that varied the feature set and the network's width and depth. Each configuration was trained with three different random seeds to smooth out the effect of random weight initialisation, and scored on its average validation ROC-AUC and F1.

- Adam optimiser with a learning rate of 0.001, and binary cross-entropy loss
- Class weighting to correct for the imbalance between churners and non-churners (~26.6% churn rate)
- Early stopping on validation AUC (patience of 15 epochs) to avoid overtraining
- Learning-rate reduction on plateau, to fine-tune convergence in later epochs

The selection rule was fixed in advance: rank all configurations by mean validation ROC-AUC, and among any that fall within one standard deviation of the top score (i.e., statistically indistinguishable), choose the one with the highest F1 score — since correctly identifying churners is the true business objective. The 64-32-16 architecture using the full feature set met this rule and was carried forward as the final design, going on to achieve a test-set ROC-AUC of 0.8172 and F1 score of 0.5944.